

EXPERIMENT 10

```
// Abstract class abstract class BankAccount {    private int accountNumber;    private String accountHolderName;
private double balance;
```

```
// Constructor
```

```
BankAccount(int accountNumber, String accountHolderName, double balance) {    this.accountNumber =
accountNumber;    this.accountHolderName = accountHolderName;    this.balance = balance;
}
```

```
// Getters    public int getAccountNumber() {        return accountNumber;
}
```

```
public String getAccountHolderName() {        return accountHolderName;
}
```

```
public double getBalance() {        return balance;
}
```

```
// Setter
```

```
public void setBalance(double balance) {        this.balance = balance;
}
```

```
// Deposit Method    void deposit(double amount) {        balance += amount;
    System.out.println("Amount Deposited: " + amount);
}
```

```
// Display Method    void displayDetails() {
    System.out.println("Account Number: " + accountNumber);
    System.out.println("Account Holder Name: " + accountHolderName);
    System.out.println("Balance: " + balance);
}
```

```
// Abstract Method    abstract void calculateInterest();
```

```

}

// SavingsAccount
class SavingsAccount extends BankAccount {

    SavingsAccount(int accountNumber, String accountHolderName, double balance) {        super(accountNumber,
accountHolderName, balance);

    }

    void calculateInterest() {

        double interest = getBalance() * 0.05;

        System.out.println("Savings Account Interest: " + interest);

    }

}

// CurrentAccount
class CurrentAccount extends BankAccount {

    CurrentAccount(int accountNumber, String accountHolderName, double balance) {        super(accountNumber,
accountHolderName, balance);

    }

    void calculateInterest() {

        double interest = getBalance() * 0.02;

        System.out.println("Current Account Interest: " + interest);

    }

}

// Main Class public class Exp10 {    public static void main(String[] args) {

    SavingsAccount sa = new SavingsAccount(101, "Rahul", 10000);

    CurrentAccount ca = new CurrentAccount(102, "Anita", 20000);

    System.out.println("----- Savings Account -----");    sa.deposit(2000);    sa.displayDetails();
sa.calculateInterest();

    System.out.println();

```

```
        System.out.println("----- Current Account -----");    ca.deposit(3000);    ca.displayDetails();
ca.calculateInterest();
    }
}
```

```
Output - Exp10 (run) ×
----- Savings Account -----
Amount Deposited: 2000.0
Account Number: 101
Account Holder Name: Rahul
Balance: 12000.0
Savings Account Interest: 600.0

----- Current Account -----
Amount Deposited: 3000.0
Account Number: 102
Account Holder Name: Anita
Balance: 23000.0
Current Account Interest: 460.0
```

EXEPERIMENT 11

```
<project xmlns="http://maven.apache.org/POM/4.0.0"      xmlns:xsi="http://www.w3.org/2001/XMLSchema-  
instance"      xsi:schemaLocation="http://maven.apache.org/POM/4.0.0  
https://maven.apache.org/xsd/maven-4.0.0.xsd">
```

```
<modelVersion>4.0.0</modelVersion>
```

```
<groupId>com.example</groupId>
```

```
<artifactId>student-rest-api</artifactId>
```

```
<version>0.0.1-SNAPSHOT</version>
```

```
<parent>
```

```
    <groupId>org.springframework.boot</groupId>
```

```
    <artifactId>spring-boot-starter-parent</artifactId>
```

```
    <version>3.2.5</version>
```

```
</parent>
```

```
<properties>
```

```
    <java.version>17</java.version>
```

```
</properties>
```

```
<dependencies>
```

```
    <!-- Spring Web Dependency -->
```

```
    <dependency>
```

```
        <groupId>org.springframework.boot</groupId>
```

```
        <artifactId>spring-boot-starter-web</artifactId>
```

```
    </dependency>
```

```
</dependencies>
```

```
<build>
```

```
    <plugins>
```

```
        <plugin>
```

```
            <groupId>org.springframework.boot</groupId>
```

```
        <artifactId>spring-boot-maven-plugin</artifactId>
    </plugin>
</plugins>
</build>

</project>
```

```
package com.example.student;
```

```
// Model class representing Student entity public class Student {
```

```
    private int id;    private String name;    private String course;
```

```
    // Default constructor    public Student() {
    }
```

```
    // Parameterized constructor    public Student(int id, String name, String course) {
        this.id = id;
        this.name = name;    this.course = course;
    }
```

```
    // Getter and Setter methods    public int getId() {    return id;
    }
```

```
    public void setId(int id) {    this.id = id;
    }
```

```
    public String getName() {    return name;
    }
```

```
    public void setName(String name) {    this.name = name;
    }
```

```

    public String getCourse() {    return course;
    }

    public void setCourse(String course) {    this.course = course;
    }
}

package com.example.student;

import org.springframework.http.HttpStatus; import org.springframework.http.ResponseEntity; import
org.springframework.web.bind.annotation.*;

// REST Controller
@RestController
@RequestMapping("/") public class StudentController {

    private Student student;

    // Welcome endpoint    @GetMapping    public String welcomeMessage() {    return "Welcome to Student
Management REST API";
    }

    // GET student details    @GetMapping("/student")    public ResponseEntity<Student> getStudent() {    if
(student == null) {
        return new ResponseEntity<>(HttpStatus.NOT_FOUND);
    }

    return new ResponseEntity<>(student, HttpStatus.OK);
    }

    // POST student data    @PostMapping("/student")    public ResponseEntity<Student>
createStudent(@RequestBody Student student) {

        this.student = student;    return new ResponseEntity<>(student, HttpStatus.CREATED);
    }
}

```

```
package com.example.student;
```

```
import org.springframework.boot.SpringApplication; import  
org.springframework.boot.autoconfigure.SpringBootApplication;
```

```
// Main Spring Boot Application @SpringBootApplication public class StudentRestApiApplication {
```

```
    public static void main(String[] args) {        SpringApplication.run(StudentRestApiApplication.class, args);  
    }  
}
```

EXEPERIMENT 12

```
<project xmlns="http://maven.apache.org/POM/4.0.0"      xmlns:xsi="http://www.w3.org/2001/XMLSchema-  
instance"      xsi:schemaLocation="http://maven.apache.org/POM/4.0.0  
https://maven.apache.org/xsd/maven-4.0.0.xsd">
```

```
<modelVersion>4.0.0</modelVersion>  <groupId>com.example</groupId>
```

```
<artifactId>employeeapi</artifactId>
```

```
<version>0.0.1-SNAPSHOT</version>
```

```
<name>employeeapi</name>
```

```
<parent>
```

```
  <groupId>org.springframework.boot</groupId>
```

```
  <artifactId>spring-boot-starter-parent</artifactId>
```

```
  <version>3.2.5</version>
```

```
</parent>
```

```
<properties>
```

```
  <java.version>17</java.version>
```

```
</properties>
```

```
<dependencies>
```

```
  <!-- Spring Web -->
```

```
  <dependency>
```

```
    <groupId>org.springframework.boot</groupId>
```

```
    <artifactId>spring-boot-starter-web</artifactId>
```

```
  </dependency>
```

```
  <!-- Spring Data JPA -->
```

```
  <dependency>
```

```
    <groupId>org.springframework.boot</groupId>
```

```
    <artifactId>spring-boot-starter-data-jpa</artifactId>
```

```
  </dependency>
```



```

<!-- H2 Database -->
<dependency>
  <groupId>com.h2database</groupId>
  <artifactId>h2</artifactId>
  <scope>runtime</scope>
</dependency>
</dependencies>

<build>
  <plugins>
    <plugin>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-maven-plugin</artifactId>
    </plugin>
  </plugins>
</build>
</project>

```

Main class package com.example.employeeapi;

```

import org.springframework.boot.SpringApplication; import
org.springframework.boot.autoconfigure.SpringBootApplication;

```

```

@SpringBootApplication public class EmployeeApiApplication {   public static void main(String[] args) {
SpringApplication.run(EmployeeApiApplication.class, args);
  }
}

```

Employee class package com.example.employeeapi.entity;

```

import jakarta.persistence.Entity; import jakarta.persistence.Id;

```

```

@Entity public class Employee {

```

```
@Id private int id; private String name; private String department; private double salary;

public Employee() {
}

public Employee(int id, String name, String department, double salary) { this.id = id;
    this.name = name; this.department = department; this.salary = salary;
}

public int getId() { return id;
}

public void setId(int id) { this.id = id;
}

public String getName() { return name;
}

public void setName(String name) { this.name = name;
}

public String getDepartment() { return department;
}

public void setDepartment(String department) { this.department = department;
}

public double getSalary() { return salary;
}

public void setSalary(double salary) { this.salary = salary;
}
}
```

```
package com.example.employeeapi.repository;
```

```
import org.springframework.data.jpa.repository.JpaRepository; import com.example.employeeapi.entity.Employee;
```

```
public interface EmployeeRepository extends JpaRepository<Employee, Integer> {  
}
```

```
Employee service package com.example.employeeapi.service;
```

```
import java.util.List; import org.springframework.beans.factory.annotation.Autowired; import  
org.springframework.stereotype.Service; import com.example.employeeapi.entity.Employee; import  
com.example.employeeapi.repository.EmployeeRepository;
```

```
@Service public class EmployeeService {    @Autowired    private EmployeeRepository repository;
```

```
    // Add Employee    public Employee saveEmployee(Employee employee) {        return repository.save(employee);  
    }
```

```
    // Get All Employees    public List<Employee> getAllEmployees() {        return repository.findAll();  
    }
```

```
    // Get Employee By ID    public Employee getEmployeeById(int id) {        return repository.findById(id).orElse(null);  
    }
```

```
    // Update Employee
```

```
    public Employee updateEmployee(int id, Employee employeeDetails) {
```

```
        Employee employee = repository.findById(id).orElse(null);
```

```
        if (employee != null) {            employee.setName(employeeDetails.getName());  
employee.setDepartment(employeeDetails.getDepartment());  
employee.setSalary(employeeDetails.getSalary());            return repository.save(employee);  
        }
```

```

        return null;
    }

    // Delete Employee
    public String deleteEmployee(int id) {
        repository.deleteById(id);
        return "Employee
deleted successfully!";
    }
}

```

Employee controller package com.example.employeeapi.controller;

```

import java.util.List; import org.springframework.beans.factory.annotation.Autowired; import
org.springframework.web.bind.annotation.*;

```

```

import com.example.employeeapi.entity.Employee; import com.example.employeeapi.service.EmployeeService;

```

```

@RestController

```

```

@RequestMapping("/employees") public class EmployeeController {

```

```

    @Autowired private EmployeeService service; // POST: Add Employee
    @PostMapping public Employee
addEmployee(@RequestBody Employee employee) {
        return service.saveEmployee(employee);
    }

```

```

    // GET: All Employees
    @GetMapping public List<Employee> getAllEmployees() {
        return
service.getAllEmployees();
    }

```

```

    // GET: Employee By ID
    @GetMapping("/{id}") public Employee getEmployeeById(@PathVariable int id) {
return service.getEmployeeById(id);
    }

```

```

    // PUT: Update Employee
    @PutMapping("/{id}") public Employee updateEmployee(@PathVariable int id,
@RequestBody Employee employee) {
        return service.updateEmployee(id, employee);
    }

```

```

    // DELETE: Delete Employee
    @DeleteMapping("/{id}") public String deleteEmployee(@PathVariable int id) {
return service.deleteEmployee(id); } }

```